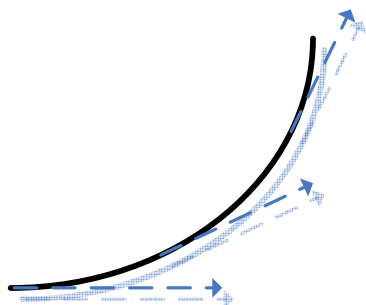


§4-3 + §4-5: 1st & 2nd Derivatives and Curve Sketching

Homework : §4-3 1,5,15,17,19,41,44,47,49,63,68,75

§4-5 3,17,45,61

1. (i) $f'(x) > 0$ on $(a,b) \Rightarrow f$ is increasing on (a,b) .
- (ii) $f'(x) < 0$ on $(a,b) \Rightarrow f$ is decreasing on (a,b)
2. (i)

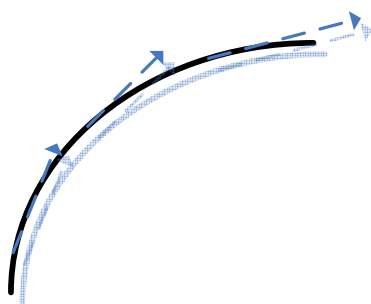


concave up (C.U.) on $(a,b) \Leftrightarrow$ 圖形在切線上方.

上方 $\Leftarrow f'$ is increasing on $(a,b) \Leftarrow f'' > 0$ on (a,b) .

(註: 完整的證明可由 2 階 Taylor's 展開式來證明。)

(ii)



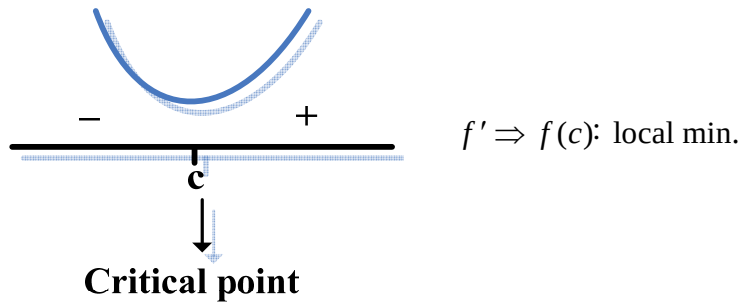
concave down (C.D.) on $(a,b) \Leftrightarrow$ 圖形在切線下方.

下方 $\Leftarrow f'$ is decreasing on $(a,b) \Leftarrow f'' < 0$ on (a,b) .

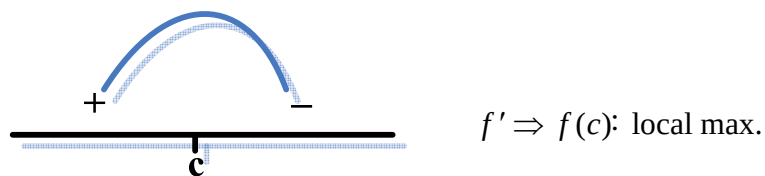
3. Inflection point: the point at which concavity is reversed.

4. 1st Derivative Test

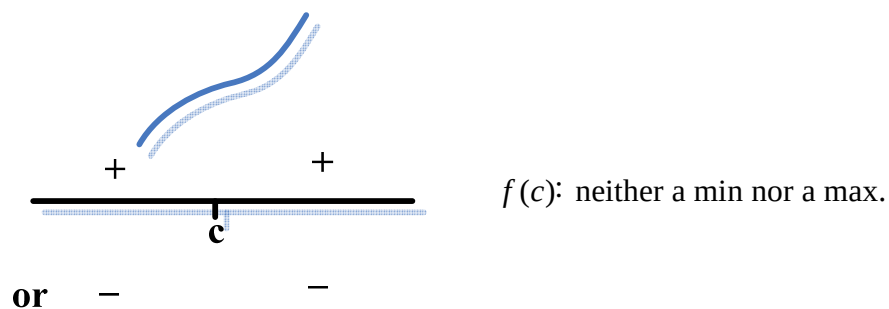
(i)



(ii)



(iii)



5. 2nd Derivative Test

c : critical point

$$f''(c) > 0 \quad f(c) : \text{local min } (\because \text{在 } c \text{ 的附近 } f \text{ C.U.})$$

$$f''(c) < 0 \quad f(c) : \text{local max } (\because \text{在 } c \text{ 的附近 } f \text{ C.D.})$$

$$f''(c) = 0 \quad \text{The test fails}$$

Example : $f(x) = x^3$, $f''(0) = 0$. $f(0)$ is neither a max. nor a min.

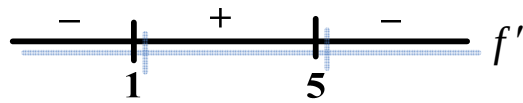
6. Curve Sketching

- (i) Intercepts (ii) Symmetry (iii) Asymptotes
 (iv) Local extrema (v) Inflection points

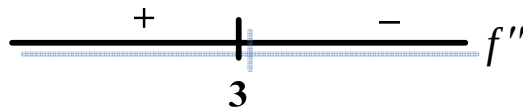
Example 1 : $f(x) = 2 - 15x + 9x^2 - x^3$

Solution : $f(x) = 2 - 15x + 9x^2 - x^3 = -(x-2)(x^2 - 7x + 1)$

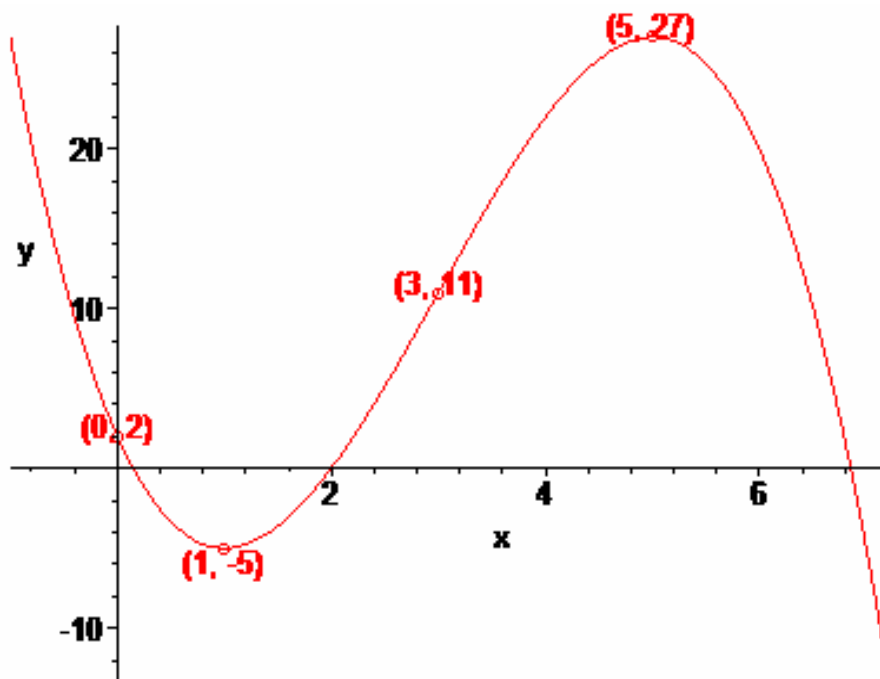
$$f'(x) = -15 + 18x - 3x^2 = -3(x^2 - 6x + 5) = -3(x-5)(x-1).$$



$$f''(x) = 18 - 6x = 6(3 - x)$$



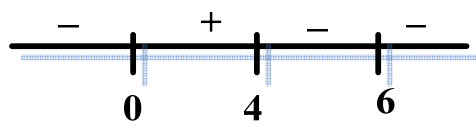
x	0	1	3	5
y	2	-5	11	27

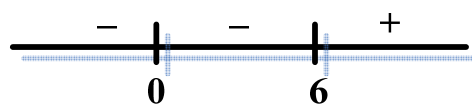


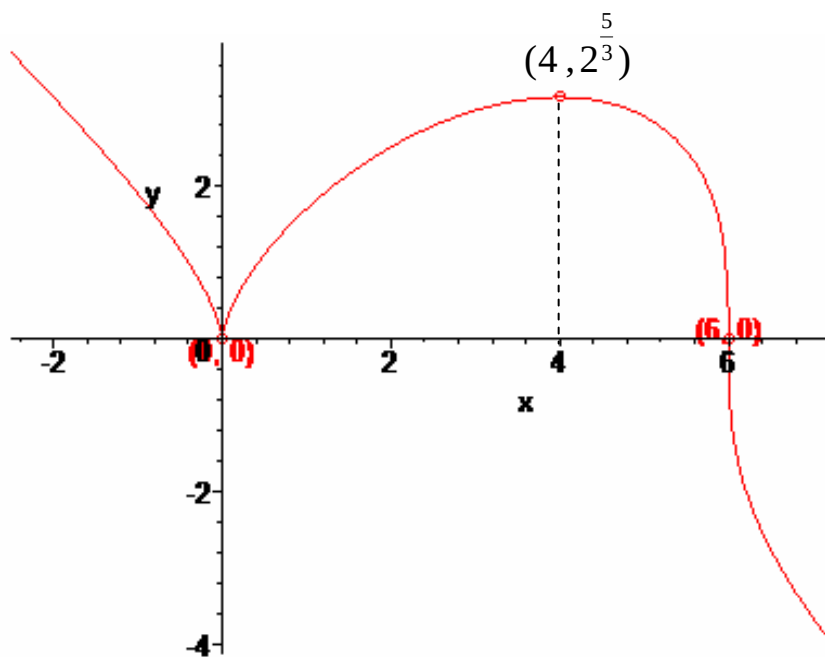
Example 2 : $f(x) = x^{\frac{2}{3}}(6-x)^{\frac{1}{3}}$

Solution :
$$\begin{aligned} f'(x) &= \frac{2}{3}x^{-\frac{1}{3}}(6-x)^{\frac{1}{3}} - \frac{1}{3}x^{\frac{2}{3}}(6-x)^{-\frac{2}{3}} \\ &= \frac{1}{3}x^{-\frac{1}{3}}(6-x)^{-\frac{2}{3}}(2(6-x)-x) \\ &= x^{-\frac{1}{3}}(6-x)^{-\frac{2}{3}}(4-x). \end{aligned}$$

$$\begin{aligned} f''(x) &= -\frac{1}{3}x^{-\frac{4}{3}}(6-x)^{-\frac{2}{3}}(4-x) + \frac{2}{3}x^{-\frac{1}{3}}(6-x)^{-\frac{5}{3}}(4-x) - x^{-\frac{1}{3}}(6-x)^{-\frac{2}{3}} \\ &= \frac{1}{3}x^{-\frac{4}{3}}(6-x)^{-\frac{5}{3}}(-(6-x)(4-x) + 2x(4-x) - 3x(6-x)) \\ &= \frac{-8}{x^{\frac{4}{3}}(6-x)^{\frac{5}{3}}}. \end{aligned}$$


 $f' \Rightarrow f(0) = 0$ is a local min.
 $f(4) = 2^{\frac{5}{3}}$ is a local max.


 $f'' \Rightarrow (6, 0)$ is an inflection point.



Example 3 : $f(x) = xe^{-x}$.

Example 4 : $f(x) = \frac{x^3}{x^2 + 1}$.

Solution : $y = f(x) = \frac{x^3}{x^2 + 1} = x - \frac{x}{x^2 + 1}$

Slant Asymptotic. $y = x$

$$y = x - \frac{x}{x^2 + 1}$$

$$\begin{array}{c} + \quad + \\ \hline 0 \end{array} f' \quad f'(x) = \frac{x^2(x^2 + 3)}{(x^2 + 1)^2}$$

$$\begin{array}{c} + \quad - \quad + \quad - \\ \hline -\sqrt{3} \quad 0 \quad \sqrt{3} \end{array} f'' \quad f''(x) = \frac{-2x(x^2 - 3)(x^2 + 1)}{(x^2 + 1)^4}$$

x	0	$\sqrt{3}$	$-\sqrt{3}$
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y	0	$\frac{3\sqrt{3}}{4}$	$-\frac{3\sqrt{3}}{4}$
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Symmetry：原點(0,0)

